EES 102: Introduction to Environmental Sciences

Instructors – Prof. Baerbel Sinha and Dr. Sunil A. Patil

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Outline of Course Contents

- History of the Universe and Earth's Geosphere, Atmosphere, Biosphere, and the Evolution of Life
- Renewable and non-renewable sources of energy, Fossil fuels, and Biofuels
- Earth's Climate: Radiation balance, Albedo: Particles and Clouds, Greenhouse Effect, International Agreements on Greenhouse Gases, Global warming and Climate change, Impacts of climate change on the Indian environment
- Air quality determinants and criteria for air pollutants and their effects; Air pollution control

Outline of Course Content (Contd)

- Environmental Health & Toxicology; Toxic Chemicals, Acute and Chronic Toxicity, Persistent Organic Pollutants in the Environment

- Natural Biogeochemical Cycles and the influence of Anthropogenic perturbations on key element cycles

- Soil and Land use: Impact of soil loss and land cover on biogeochemical cycle, Fertilizers and Green Revolution, Agricultural practices and environmental footprints, The future of food

- Biodiversity, Problems and issues in biodiversity and forestry, Conservation and utilization of biodiversity, Biomes, Landscapes, Ecosystems

Outline of Course Content

- Hydrological cycle, Water resource conservation, Rain water harvesting methods, Regulation of Water Quality, Water Treatment, Water purification methods: pros and cons, Water footprint of consumer products

- Managing our waste: Wastewater treatment, Solid Waste Management, Waste as a resource through 3Rs

- Sustainable development, Case Studies of Environmental Pollution Episodes and Successful Interventions, Glimpses of Field Work, Creating Livable/Sustainable Cities

Outline of Course Content (Contd)

Recommended Books and Learning Material:

- J. Girard, Principles of Environmental Chemistry, Jones & Bartlett Learning, 2009/2013
- J. H. Withgott & M. Laposata, Environment: The Science Behind the Stories, 7th Ed., Pearson, 2021
- W. Schlesinger & E. S. Bernardt, Biogeochemistry: An Analysis of Global Change, Academic Press, 2020
- M. Jacobson, R. J. Charlson, H. Rodhe, G. H. Orians, Earth System Science, From Biogeochemical Cycles to Global Changes, Volume 72: International Geophysics, Academic Press, 2020
- S. Karr, Environmental Science for a Changing World, Scientific American and Macmillan Learning, 4th Ed., 2021
- D.B. Botkin and E.A. Keller, Earth As a Living Planet, John Wiley & Sons, 8th Ed., 2010
- S. E. Manahan, Environmental Chemistry, CRC Press, 9th Ed., 2009
- Down to Earth magazine and Select Journal Articles

Evaluation/Grading Criteria

Total Marks	**:**	**100**
Mid Sem Exam I	**:**	**20**
Mid Sem exam II	**:**	**20**
End Sem Exam	**:**	**50**
Written Quiz	**:**	**06 (averaged over all quizzes)**
Attendance	**:**	**04**

(Policy: >90 % 4/4; 80-90% 3/4; 70-80% 2/4; 60-70% 1/4; <60% 0/4)

Grading: Relative

Important Note: Students having <u>less than 50% attendance</u> will not be permitted to take the end-sem exam

Important points

- Please dedicate a separate notebook to take class notes
- Classroom notes are most important from the learning and exam point of view – select material will be shared via Moodle/another platform.
- Enroll yourself on Moodle for accessing additional helpful aids

- EES 102 is a Mandatory course which will be useful for making an informed decision on your pre-majors and options of BS-MS degree: Majors in Geology Or Environmental Sciences
- Also offered by EES dept are Minors in Atmospheric Science/Environmental Science/Earth Science

Instructor's background

- **Prof. Dr. Baerbel Sinha**
  Trained as an Archaeologist who is an Atmospheric chemist

https://www.iisermohali.ac.in/faculty/ees/bsinha

- **Dr. Sunil A. Patil**
  Trained as a microbiologist who works in Environmental microbiology and biotechnology

https://www.iisermohali.ac.in/faculty/ees/sunil

Teaching assistants (Postdocs/PhDs):
Dr. Sachin, Dr. Ashmita, Arpit, Krishna and Priya